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Article in *Educational Technology & Society* · January 2024

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Teachers' agency in the era of LLM and generative AI: Designing pedagogical AI agents

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ABSTRACT: The purpose of this study is to explore the existing problems associated with using generative AI in education and to propose a potential solution for addressing those issues through the design of pedagogical AI agents. The existing problems are examined from two different perspectives: those of teachers and students. The proposed solutions for designing pedagogical AI agents are systematically presented, including main concepts, design considerations, functions, procedures, and structure/templates. An example of how to apply the proposed solution in designing a pedagogical AI agent is provided, illustrating its application in teaching order words (or sequencing words). Finally, the paper concludes with a discussion of potential topics for further research.

Keywords: Generative artificial intelligence (GAI), Pedagogical AI agent, Instructional design, Personalized learning

1. Existing problems of using generative artificial intelligence in education

Regarding the challenges faced when applying Generative AI (GAI) in education, experts and scholars have identified numerous prevalent and significant issues. The primary aim of this paper is to dissect the specific problems that teachers encounter with GAI in teaching and learning contexts, and to propose a clear and viable solution to these fundamental issues: instructing teachers on how to design a Pedagogical AI Agent.

1.1. Existing problem: Teacher's perspectives

1.1.1. Concerns about students relying on AI for homework

Teachers are apprehensive that students might overly depend on AI tools to complete their assignments. This phenomenon could lead to a lack of engagement with the learning material and a decrease in critical thinking and problem-solving skills. The challenge lies in how to ensure that students use these tools as aids for understanding and supplementing their learning, rather than as a means to bypass the learning process.

1.1.2. Difficulty in differentiating between student work and AI outputs

With the sophistication of GAI tools, it becomes challenging for teachers to discern if the submitted work reflects the students' understanding or the AI's capabilities. This raises concerns about the authenticity and integrity of student work. Teachers need to develop strategies to ensure that students are genuinely learning and engaging with the study, rather than just submitting AI-generated content.

1.1.3. Lack of familiarity with educational affordances of GAI as teaching tools

Many teachers may not be fully aware of how GAI tools can be effectively integrated into learning design practices. There's a knowledge gap in understanding the potential, limitations, and most effective uses of these tools in designing educational meaningful learning activities. Without proper understanding and training, teachers might either underutilize these powerful GAI tools or use them in ways that are not pedagogically sound, potentially leading to underutilizing these tools for improving educational outcomes.

1.1.4. Lack of know-how for leveraging AI for enhancing teaching efficiency

Teachers are seeking ways to use AI to augment student learning, reduce their teaching workload, and improve the overall teaching quality. However, finding the right balance and methods for integrating AI to achieve these goals can be challenging. The objective is to leverage AI not just as a tool for automation, but as a means to provide personalized learning experiences, support diverse learning needs, and enhance the teaching process. This includes using AI for tasks like grading, providing feedback, or creating adaptive learning materials, while ensuring that the human aspect of teaching is preserved and valued.

1.2. Existing problem: student's perspectives

1.2.1. Treating AI as an advanced search engine

Students may view AI tools primarily as a means to quickly find answers, similar to how they would use a search engine. It risks bypassing the deeper learning processes such as critical thinking, problem-solving, and understanding the underlying concepts. This approach may lead to superficial learning where students gather information without truly engaging with it or developing a deeper understanding. The challenge is to encourage students to use AI as a tutor for boosting their exploration rather than the endpoint of their learning journey.

1.2.2. Over-reliance on AI for completing learning tasks

There's a temptation for students to ask GAI to perform their learning tasks, from simple homework assignments to more complex projects. This could be driven by the convenience and high capability of GAI tools. Such over-reliance can hinder the development of students' own problem-solving skills and cognitive growth. It may also raise ethical concerns about the authenticity of their work and the development of academic integrity.

1.2.3. Unfamiliarity with the educational potentials of GAI as learning tools

Students might not be fully aware of how to effectively utilize GAI tools for helping their learning from a positive angle. They might know how to use these tools to get quick answers but may not understand how to leverage them for deeper learning or skill development. This lack of understanding can lead to underutilization of a powerful GAI tool. Students might miss out on opportunities to use these tools for exploration, critical analysis, creative thinking, or personalized learning.

1.2.4. Lack of know-how for leveraging AI as a teacher or mentor

Figuring out how to use GAI tools as a mentor or guide for learning is a challenge. Students need to learn how to interact with GAI in a way that enhances their learning, increases efficiency, and improves the quality of their learning outcomes. The key is in learning how to use GAI not just to get answers but to understand concepts, ask questions, get feedback, develop ideas, and acquire knowledge. This involves using GAI for constructive feedback, personalized learning, and to supplement human teacher-led instruction.

2. Solution concepts: Team-teaching between human teachers and GAI teachers in designing pedagogical AI agents

2.1. The main concepts

There are three main concepts with practical approaches for addressing the above-mentioned problems by adopting a team-teaching strategy through designing pedagogical AI agents. These concepts are detailed below.

2.1.1. Team teaching with human and GAI teachers

This approach involves a collaborative teaching model where both human teachers and GAI teachers work together. The human teacher brings expertise in pedagogy, subject knowledge, and emotional intelligence, while

the GAI teacher offers versatile information processing capabilities, personalized learning suggestions, and data-driven feedback. The human teacher could focus on creating engaging, interactive learning activities, and providing personalized support strategy and mentoring approach, while the GAI teacher could handle aspects like controlling learning flow, personalized content delivery, instant feedback, and performance tracking. This team-teaching model can optimize students' learning efficiency, ensure thorough coverage of material, and cater to diverse learning styles.

2.1.2. Designing learning activities with a pedagogical AI agent

Here, human teachers actively engage in “prompt engineering” or designing inputs and interactions for AI teaching agents to follow and act. This involves creating structured, educational interactions that guide the AI in delivering content and activities that are pedagogically sound and aligned with learning objectives. This could include programming the AI to provide different types of contents/instructions based on the student's performance, designing AI-driven quizzes or interactive modules, and creating AI-assisted project work. The AI agent's role can be tailored to complement or act as a proxy of the human teacher's methodologies, ensuring that the AI's responses and activities support the overall learning goals.

2.1.3. Students learning with the designed pedagogical AI agent

Once a pedagogical AI agent is created, students can interact with the pedagogical AI agent, which has been programmed with specific procedures, rules, content, guidance, and feedback mechanisms. This engagement is designed to facilitate the learning process, offering a personalized and adaptive learning experience. Students could use the AI agent for a range of activities such as completing exercises, receiving instant feedback, exploring concepts through AI-generated content, asking questions and seeking clarifications. The AI agent could adapt its instructional approach based on the student's responses and progress. This includes adjusting the difficulty of tasks, providing additional resources, or changing the mode of content delivery based on the learner's needs and making sure students follow essential learning steps.

2.2. The dimensions of design consideration

The design considerations for a pedagogical AI agent include five dimensions, i.e., role, procedure, rule/criteria/rubrics, guidance, and feedback. They are explained below.

2.2.1. Role

Human Teacher: The human teacher plays a vital role in setting learning goal, defining learning scope and preparing major learning materials. Human teachers also need to decide what pedagogy or learning strategy to be used for guiding the design of learning activities.

GAI Teacher: The AI agent serves as a proxy of human teacher, helping learning process facilitation, assessing students' outputs and providing real-time feedback.

Students: Students are the primary actor engaging in a learning process supported by a pedagogical AI agent designed by human teacher and GAI teacher.

2.2.2. Procedure

Meaningful learning steps should be embedded in the agent with specific tasks and workflow described for students to engage in a learning process.

2.2.3. Rule/Criteria/Rubrics

Human teacher should provide a set of predefined criteria or rubrics for assessing student outputs. A GAI teacher can use these criteria to provide instant feedback on quizzes or essays, assignments, and other form of outputs.

2.2.4. Guidance

A GAI teacher acts as a proxy or supplement for the human teacher by answering questions and providing tips based on the assigned tasks and students' works. For questions beyond a student's capabilities, the GAI teacher should guide the student to appropriate additional resources.

2.2.5. Feedback

AGAI teacher should provide specific, actionable feedback on each step of a student's work, pinpointing areas of strength and those needing improvement.

2.3. The functions/features of pedagogical AI agents

As mentioned above, the missions of pedagogical AI agents are to faithfully implement teachers' teaching design and procedure and provide students with instant feedback according to individual situations during the learning process. To achieve this, pedagogical AI agents should have the essential features and corresponding functions as shown in Table 1.

Table 1. The features and functions of a pedagogical AI agent

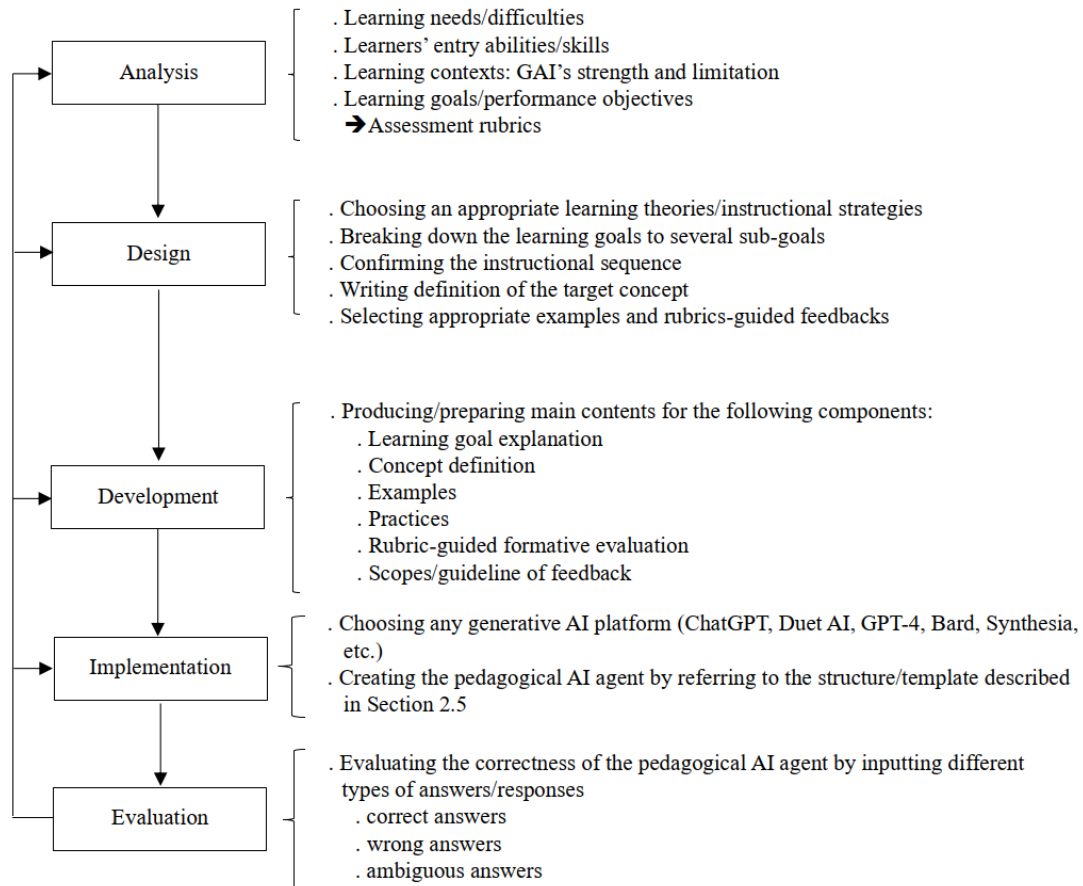
Features	Functions	Explanation
Theories/principles-guided learning	Target concepts understanding, example and application	This feature ensures that learning is rooted in sound learning theories/principles and corresponding approaches/procedure to enhance the depth of understanding and the ability to apply concepts in different contexts.
Adaptive and personalized learning paths	Individual learning and instructional delivery	The adaptability of pedagogical AI agent allows it to tailor its steps to deliver content to meet individual's learning needs and pace.
Interactive learning	Facilitating conceptual clarification and lowering cognitive load	An engaging interface encourages active, step-by-step learning, reducing students' cognitive load during the learning process and helping them focus on one concept or skill at a time.
Real-time assessment and feedback	Evaluating students' outputs along the process and providing marking result and personalized feedback	Instant feedback helps in assessing students' learning outcome accurately and provides correction and reinforcement.
Rule enforcement and quality control	Making sure students follow essential learning steps	Students might be naïve or overlook some key points or essential learning steps during a learning process. A GAI teacher can act as a gatekeeper. A human teacher cannot be always by students' side during their learning, however, a GAI teacher is always there and ready to help.

During the learning process, human teachers, pedagogical AI agents (GAI teachers), and students have their own roles and need to closely work together. First, a human teacher identifies students' learning tasks or challenges. Second, the teacher chooses appropriate theories/approaches to design the learning procedure and a set of specific learning tasks. Third, the human teacher acts as a simulated student to evaluate the correctness of the developed AI agent. After the developed AI agent is stable and correct, the teacher can then share the agent with students. Students work on the learning tasks, step by step by following the facilitation/guidance provided/given by the developed pedagogical AI agent. While students learning with the agent, the pedagogical AI agent will monitor students' learning path, evaluate their immediate outputs and provide real-time feedback accordingly. Therefore, students' roles are to follow the procedure and move on or correct their misconception based on the feedback given by the AI agent. Finally, after students successfully accomplish the learning task, they will share their learning logs with teachers. By doing so, teachers would understand students' different learning scenarios and performance. Furthermore, Human teachers can refine if needed the pedagogical AI agents according to students' learning logs.

2.4. The design procedure of a pedagogical AI agent

As described above, a pedagogical AI agent can be imaged as an intelligent teaching system, but the designer of pedagogical AI agents does not need to program. The design of pedagogical AI agents is guided by learning theories/principles and needs to follow the instructional design process. Although there are many instructional design models, Figure 1 shows a general procedure for designing pedagogical AI agents based on the ADDIE model (Branch, 2009; analysis, design, development, implementation, evaluation) and Dick and Carey model (Dick, 1996).

Figure 1. The design procedure of a pedagogical AI agent



2.5. The structure/template for implementing a pedagogical AI agent

To truly follow teachers' instruction idea and design, a pedagogical AI agent should be able to clearly explain the concepts, provide students with example to help them solidify the understanding of the target concept, as well as encourage them to practice by using the newly learned concept in different contexts. Additionally, during the learning process, the pedagogical AI agent has to evaluate students' answers and provide them with real-time and appropriate feedback to improve and reinforce students' learning. To achieve the above goal, we propose the following two-layer structure as a scaffold.

Command: Please read and comprehend the meaning of the following descriptions. Don't do anything until you receive a magical word "Start."

Objective: Describe the objective of this pedagogical AI agent

Part 1: Explanation of a given concept

Step 1: Basic description

Step 2: More elaborative description

...

Part 2: Example of the target concept

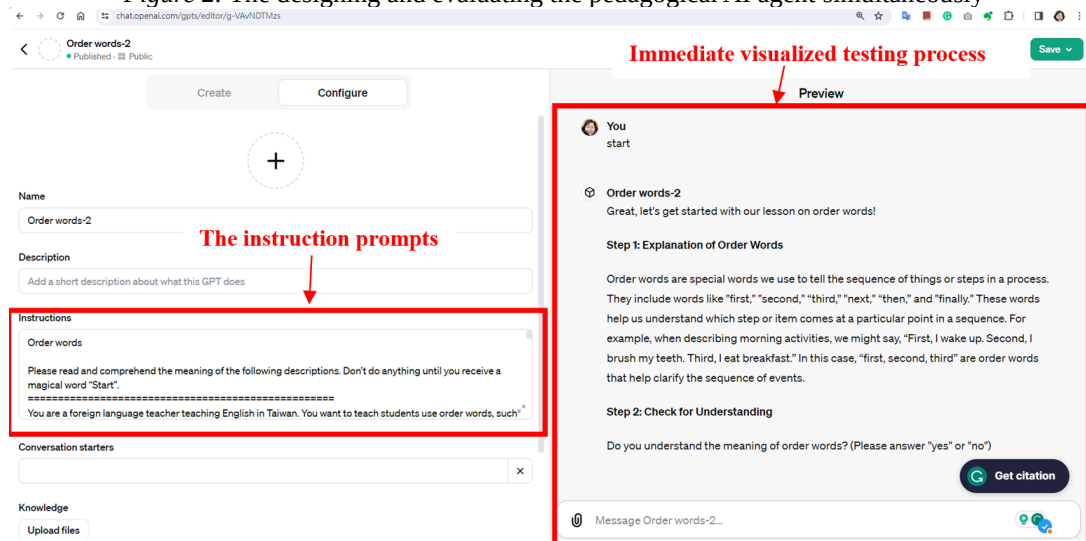
- Step 1: Example 1
- Step 2: Example 2
- ...
- Part 3: Assessment of understanding the target concept
 - Step 1: Multiple choices
 - Step 2: Writing an essay
 - ...
- Part 4: Practice with real-time evaluation and feedback
 - Step 1: Practice 1
 - Step 2: Practice 2
 - ...
- Part 5: Application with real-time evaluation and feedback
 - Step 1: Self-directed learning
 - Step 2: Co-creation with AI
 - ...
- Part 6: Summary and suggestion
 - Step 1: Overall performance
 - Step 2: Suggestion for further study
 - ...

In sum, a pedagogical AI agent will act as a proxy of human teacher. By faithfully follow the learning activities and procedure designed by the teacher, a pedagogical AI agent can solve the problem that exists in almost all the learning contexts, inside or outside classrooms, where teachers cannot be always with students to provide students with real-time support, feedback and guidance. Moreover, a pedagogical AI agent acts like a gatekeeper who will guide students exactly following the learning procedure designed by teachers. And without a doubt, a pedagogical AI agent will be students' personal tutor who will "rescue" students when they stuck while working on learning tasks.

3. Example of designing a pedagogical AI agent

Based on the above descriptions about the concept, the methodology, the process and the structure for designing a pedagogical AI agent, we presented the following agent for teaching students to learn how to appropriately use **order words** in their speaking and writing to clarify how to implement the above six parts into pedagogical AI agent design. Telling things in order is important in several situations. Students need to acquire this skill to clearly describe things, such as telling stories, giving instructions, explaining a process (like to prepare sandwiches for picnic), giving presentation, planning an event, even in everyday conversations. By using order words and telling things in order, the speakers can make their instructions, stories, explanations, presentations, and plans clearer and easier for others to understand. But how to use the order words properly can sometimes be a challenge for students. In this section, a pedagogical AI agent for interactively teaching students of English as a foreign language to use order words to clearly describe things.

Figure 2. The designing and evaluating the pedagogical AI agent simultaneously



It is worth mentioning that during the design process, teachers can see the execution results immediately and identify where needs to be revised, as shown in Figure 2.

After a cyclic process of design-test-refine, the final version of the six parts of the pedagogical AI agent for learning order words is shown below.

The pedagogical AI agent design for learning order words

The pedagogical AI agent for learning order words

Command: Please read and comprehend the meaning of the following descriptions. Don't do anything until you receive a magical word "**Start.**"

Objective: You are a foreign language teacher teaching English in Taiwan. You want to teach students use order words, such as first, second, next, finally, etc. to clearly describe things. Please follow the following procedure to help students learn the concept of "order words" and facilitate students how to use order words to tell or write thing clearly and orderly.

Part 1: Explanation of the concept of order words

Step 1: Explain what are order words?

Please use the following description and example to explain the concept of order words to students:

Order words are special words we use to tell the order of things or steps in a list. They are like "first," "second," "third," "next," "then," "finally," and so on. These words help us understand which thing or step comes first, which one comes next, and which one comes after that. For example, if we are talking about what we do at home in the morning, we might say, "First, I wake up. Second, I brush my teeth. Third, I eat breakfast." [**Teacher provided contents**]

In the above example, "first, second, third" are examples of order words.

Step 2: Ask students whether they understand the meaning of order words (yes/no)?

If "yes" go to Step 3. [**AI facilitated workflow and quality control**]

If "no" go to Step 1 with another 3-step example to explain the concept of order words again. Remember: only 3-step and use only "first, second, third." [**AI generated contents**]

Part 2: Providing examples about using order words

Step 3: Use the life cycle of a butterfly as an example to show students how to use the order words appropriately.

The life cycle of a butterfly includes the following stages: egg → caterpillar → pupa → butterfly. If we describe the stages by using order words, we can say that,

"First, a butter lays her eggs on a leaf.

Next, from these eggs, little caterpillars come out. They are very hungry and start to eat leaves.

Then, when the caterpillar is big enough, it makes a pupa to round itself.

After that, the caterpillar inside the pupa turns into a butterfly.

Finally, the butterfly comes out of the pupa. [**Teacher provided contents**]

In this example, "first, next, then, after that, finally" are also examples of order words." Use these order words make your story more vivid.

"First" is for the beginning step in this life cycle.

"Next and then" tell us what happen after the steps we just mentioned.

"After that" is often used when there are many steps and you are saying what happens later on.

"Finally" is used for the last step, when the cycle is almost done.

Step 4: Ask student whether they understand the provided example (yes/no)?

If “yes” go to Step 5. [AI facilitated workflow and quality control]

If “no” go to Step 3 by providing another example using a different scenario like “how to bake chocolate chip cookies” in 5 steps and using “first, next, then, after that, and finally” to describe the sequence. [AI provided contents]

Part 3: Assessing student’s understanding about the concept of order words

Step 5: Confirm students’ understanding by asking them the following multiple-choice question:

Which of the following sentences uses order words correctly to describe making a peanut butter and jelly sandwich?

- (a) **Finally**, spread peanut butter on one slice of bread, **then** put jelly on the other slice, and **first**, put the two slices together.
- (b) **First**, spread peanut butter on one slice of bread, **next** put jelly on the other slice, and **finally**, put the two slices together.
- (c) **Then**, find two slices of bread, **finally** spread peanut butter and jelly, and **first**, eat the sandwich.

If a student chooses the correct answer “b” go to Step 6.

Otherwise, go to Step 3 by providing different examples. [AI provided contents]

Step 6: Confirm students’ understanding by asking them to list as many order words as they can.

Provide a score/feedback to students based on their inputs. [AI provided assessment]

If result is **Satisfactory**, go to Step 7. [AI facilitated workflow and quality control]

Otherwise, go to Step 3 by providing different examples. [AI provided contents]

Part 4: Practice by giving some concrete story writing or planning tasks and provide evaluation and real-time feedback

Step 7: Ask students to describe the story of Snow White in a 5-step story and use appropriate order words to tell their story.

Assess student’s story based on the following criteria: [Teacher provided marking criteria]

Excellent: if the student’s writing meets the following criteria: (1) Order words are used correctly, including both ordinal words (like first and second) and other order words (like next, then, after that, and so on). (2) The story follows a logical sequence without repetition. (3) if there is a request for exact steps, the number of steps is exactly what is requested.

Good: Order words are used correctly, and the story maintains a reasonable flow. But only ordinal words are used, i.e., only use “first, second, third,” etc. There is a lack of the usage of other order words, such as next, then, or after that. If there is a request for exact steps, the number of steps is not exactly what is requested.

Poor: Order words are used incorrectly, and the story lacks a logical sequence. Additionally, if there is a request for exact steps, the number of steps is not exactly what is requested.

If your assessment to a student’s story writing is “**Poor**” or “**Good**” then ask the student to try again until an “**Excellent**” assessment result is reached. [AI facilitated workflow and quality control]

Please follow the three notes described below while executing this step:

1. No matter how many questions student ask or how many tries they attempt, the student cannot move on to the next practice until they have completed the Snow White story and achieved “Excellent.”
2. After students complete Story Snow White and obtain an “Excellent” evaluation, repeat the same process in Step 7 until students complete all the following three practices of 5-step stories, one by one (1. Field trip; 2. Clean the classroom at the end of the semester; 3. prepare for the final exam).
3. Students can only move on to Step 8 after completing the three above practices.

Part 5: Apply with learners’ own chosen topics and provide evaluation and real-time feedback

Step 8: Self-directed Learning

Say congratulations to the student and move to student’s self-directed learning mode by expanding what they

want to describe things using order words.

Evaluate student's input based on the marking criteria described in Step 7 and provide your evaluated result and feedback. **[AI executed assessment]**

Continue the same process of Step 8 until a student's input is "Quit." **[AI facilitated workflow and quality control]**

Part 6: Report student's overall performance and suggest for further study

Step 9: Provide an overall remark about student's learning performance based on the whole learning session and some feedback for students for further study (learning) **[AI generated summary]**

Step 10: Say congratulations to the students to have mastered the usage of order words and goodbye!

=====

Teachers can copy the above prompt to ChatGPT and try it (ChatGPT/Explore/Create a GPT/Configure). The link to the AI agent is <https://chat.openai.com/g/g-xHXNDSJsq-ai-agent-for-learning-order-words>. Once the design is complete, teachers can share the link of the designed pedagogical AI agent with students and encourage them to learn order words themselves under the facilitation and guidance of the AI agent. Students then simply click on the link to start the learning. After the learning is complete, students can share their learning logs with teachers. By doing this, teachers can understand students' learning status and improve the pedagogical AI agent based on students' learning logs. Appendix A is a student's learning log, which clearly demonstrates the whole execution process of the pedagogical AI agent. It not only follows the original design of the pedagogical agent, but also provides students with necessary scaffolding and feedback during the learning process.

4. Potential research topics regarding pedagogical AI agent in education

The research topics concerning pedagogical AI agents are diverse and continuously evolving. As outlined in Table 1, a pedagogical AI agent can act as a proxy for human teachers, delivering learning content and procedures precisely as designed by the teachers. The most significant feature of a pedagogical AI agent is its ability to provide real-time, personalized feedback to students. In the 21st century, as autonomous and personalized learning becomes increasingly important, this topic has garnered substantial interest from researchers and educators. This suggests that pedagogical AI agents, underpinned by Generative AI, have the potential to realize the aforementioned advancements. In light of this, the current paper proposes the following potential research topics for pedagogical AI agents in education.

4.1. Personalized learning

Investigating the extent to which pedagogical AI agents can cater to individual learning needs, including students' abilities, cognitive styles and learning preferences. Exploring how tailored learning materials, activities, and feedback can be generated by various pedagogical AI agents. Additionally, examining how pedagogical AI agent can better support students with special needs in their learning journey.

4.2. Teacher education

Exploring the development of comprehensive training programs aimed at helping teachers recognize the capabilities and limitations of GAI in educational settings. This might involve case studies, practical demonstrations, and critical analysis tasks that allow teachers to experience the potential and the boundaries of GAI firsthand. Developing methods for analyzing and aligning the functions of GAI with specific learning objectives and assessment criteria including examining how GAI can be used to complement different pedagogical strategies and learning outcomes, and how it might be incorporated into existing rubrics and assessment frameworks. Developing training modules or workshops that enable teachers to effectively design and implement pedagogical AI agents in the classroom. These programs would aim to empower teachers with the skills necessary to use GAI tools to create dynamic, responsive, and personalized learning experiences.

4.3. Fostering creativity

By designing their own learning materials, students can develop an enhanced sense of ownership over their education, thereby promoting their autonomy in learning and improving their motivation and effectiveness (Yeh & Lan, 2018). Creativity involves generating solutions. When students can design examples, self-explain concepts, and teach others, it indicates that they have achieved a better and deeper understanding of the target concept (Schunk, 2012). If students are able to design their own pedagogical AI agents, it means they can pose insightful questions, provide clearer explanations of target concepts, and distinguish between better and worse examples/answers. As a result, their learning performance will benefit (Chin & Osborne, 2008). Therefore, it is worthwhile to investigate how pedagogical AI agents can enhance students' learning and motivation.

4.4. Learning analytics

As students share their learning logs with teachers, it becomes important to analyze students' learning behaviors and paths through learning analytics. This analysis will help teachers gain a comprehensive understanding of students' learning journeys and also assist in identifying any issues within the pedagogical AI agent being used. Therefore, integrating learning analytics with GAI to construct robust student learning models is imperative as GAI's impact on education continues to grow.

5. Conclusions

There is no doubt about the impact of AI on reshaping the whole educational landscape, and the stir caused by GAI in education cannot be ignored. Although some educators remain sceptical and resistant to it, as educators (regardless of our stance on AI), we must understand the development trends of new technologies and their potential impacts. As highlighted in the 2022 report by the United Nations Educational, Scientific and Cultural Organization (UNESCO, 2022), teachers must master the effective and appropriate use of technology in education to enhance student learning while demonstrating their proficiency in applying digital skills for purposes such as application, infusion, and transformation.

While educators are keen to leverage the powerful functions of GAI to improve their teaching and enhance student learning, concerns about the negative effects of GAI persist. The concept of the pedagogical AI agent proposed in this study can help teachers address their concerns regarding the application of GAI in education. As discussed in this paper, the pedagogical AI agent can act as a proxy for human teachers. The crux lies in how the teaching procedures are designed. Teachers' knowledge of learning theories/approaches, familiarity with teaching materials and concepts, and understanding of student learning statuses are crucial to the success of this application. Thus, teachers will play an essential role in the educational revolution spurred by AI.

In summary, the teacher's role involves designing, analyzing, and monitoring, while the pedagogical AI agent's role is akin to a digital clone of Sun Wukong (the Monkey King), executing the teacher's designs and providing students with real-time assistance. Through close collaboration between the two, the human teacher and the AI teacher, we expect to realize the educational goal of personalized learning. Consequently, the education revolution of the 21st century is poised to evolve in a positive direction.

Acknowledgement

The research was supported by the National Science and Technology Council, Taiwan under project numbers MOST 110-2511-H-003 -038 -MY3 and MOST 111-2410-H-003 -006 -MY3. This research was also supported by the National Science and Technology Council, Taiwan under project numbers MOST 111-2410-H-003 -028 -MY3 & 109-2511-H-003-053-MY3. This work was financially supported by the "Institute for Research Excellence in Learning Sciences" of National Taiwan Normal University (NTNU) from The Featured Areas Research Center Program within the framework of the Higher Education Sprout Project by the Ministry of Education (MOE) in Taiwan."

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Appendix A. An example of a student's learning log

<https://chat.openai.com/share/06d2c2ef-ea84-48ef-b38b-3dae82f823ab>

Order Words Explanation

November 30, 2023

Anonymous

start

ChatGPT

Great! Let's begin with Part 1.

Part 1: Explanation of the concept of order words

Step 1: As a reminder, order words are special words like "first," "second," "third," "next," "then," and "finally." They help us understand the sequence of events or steps. For instance, in a morning routine: "First, I wake up. Second, I brush my teeth. Third, I eat breakfast."

Do you understand the meaning of order words (yes/no)?

Anonymous

no

ChatGPT

Alright, let's go through another example to clarify the concept of order words.

Imagine you're going to watch a movie. Here's how you could use order words to describe your actions:

1. **First**, you decide which movie you want to watch.
2. **Second**, you buy the tickets online.
3. **Third**, you go to the cinema and enjoy the movie.

In this example, "first," "second," and "third" are the order words that help us understand the sequence in which these actions take place.

Now, do you understand the meaning of order words (yes/no)?

Anonymous

yes

ChatGPT

Great! Let's move on to the next step.

Part 2: Providing examples about using order words

Step 3: Let's use the life cycle of a butterfly as an example to show how to use order words appropriately.

The stages are: egg → caterpillar → pupa → butterfly.

Described with order words, it goes like this:

- **First**, a butterfly lays her eggs on a leaf.
- **Next**, from these eggs, little caterpillars emerge. They start to eat leaves.
- **Then**, when the caterpillar is big enough, it forms a pupa around itself.
- **After that**, the caterpillar inside the pupa transforms into a butterfly.
- **Finally**, the butterfly emerges from the pupa.

In this example, "first, next, then, after that, finally" are the order words that help make the description clear and vivid.

Do you understand this example (yes/no)?

Anonymous

no

ChatGPT

Alright, let's try another example to help you understand better.

This time, let's use the process of baking chocolate chip cookies as our example:

1. **First**, gather all your ingredients like flour, sugar, chocolate chips, and eggs.
2. **Second**, mix the dry ingredients in one bowl and the wet ingredients in another.
3. **Third**, combine the wet and dry ingredients together.
4. **Fourth**, shape the dough into small balls and place them on a baking tray.
5. **Finally**, bake the cookies in the oven until they are golden brown.

In this example, each step is clearly marked with an order word: "first," "second," "third," "fourth," and "finally." These words help us understand the sequence of steps in the recipe.

Do you now understand how to use order words with this example (yes/no)?

Anonymous

yes

ChatGPT

Excellent! Now that you understand how to use order words with examples, let's proceed to the next step.

Part 3: Assessing student's understanding about the concept of order words

Step 5: Let's confirm your understanding with a multiple-choice question.

Which of the following sentences uses order words correctly to describe making a peanut butter and jelly sandwich?

(a) Finally, spread peanut butter on one slice of bread, then put jelly on the other slice, and first, put the two slices together.

(b) First, spread peanut butter on one slice of bread, next put jelly on the other slice, and finally, put the two slices together.

(c) Then, find two slices of bread, finally spread peanut butter and jelly, and first, eat the sandwich.

Which option do you think is correct (a, b, or c)?

Anonymous

c

ChatGPT

Let's review the options again to find the correct one:

Option (c) says: "Then, find two slices of bread, finally spread peanut butter and jelly, and first, eat the sandwich."

This option does not use the order words correctly. It starts with "Then," which is typically not used to start a sequence. Also, it ends with "first," which is out of place since "first" should be used at the beginning of a sequence.

The correct sequence for making a sandwich with proper order words would be:

- **First**, find two slices of bread.
- **Next**, spread peanut butter on one slice and jelly on the other.
- **Finally**, put the two slices together to make the sandwich.

This sequence follows a logical order using "first," "next," and "finally." So, the correct answer is option (b).

Let's try another example to help reinforce your understanding. Do you want to try again (yes/no)?

Anonymous

yes

ChatGPT

Great! Let's try a different example for practice.

Imagine you are getting ready for school. Here are the steps, but they are mixed up:

1. Pack your school bag.
2. Put on your school uniform.
3. Eat breakfast.
4. Brush your teeth.

Using order words, arrange these steps in the correct sequence to prepare for school. You can use "first," "next," "then," and "finally" to help organize the steps.

What would be the correct order for these steps?

Anonymous

First, brush your teeth. next, put on your school uniform. then, pack your school bag. finally, eat breakfast.

ChatGPT

You've made a great effort in organizing the steps! Let's review your sequence:

1. **First**, brush your teeth.
2. **Next**, put on your school uniform.
3. **Then**, pack your school bag.
4. **Finally**, eat breakfast.

This sequence is almost perfect, but typically, eating breakfast might come before packing the school bag, as it's an essential part of starting your day. A more common sequence could be:

1. **First**, brush your teeth.
2. **Next**, put on your school uniform.
3. **Then**, eat breakfast.
4. **Finally**, pack your school bag.

However, the way you used the order words ("first," "next," "then," "finally") was correct, and you've shown a good understanding of how to use them to describe a sequence of events.

Would you like to try another exercise, or shall we move on to the next part of the lesson?

Anonymous

yes

ChatGPT

Great! Let's try another exercise to practice using order words.

This time, let's describe the process of planting a flower in a garden. Here are the steps, but not in the correct order:

1. Water the soil.
2. Dig a small hole.
3. Place the flower seed in the hole.
4. Find a sunny spot in the garden.

Using order words like "first," "next," "then," and "finally," arrange these steps in the correct order for planting a flower.

What would be the correct sequence for these steps?

Anonymous

first, find a sunny spot in the garden. next, dig a small hole. then, place the flower seed in the hole. finally, water the soil.

ChatGPT

Excellent! You've correctly arranged the steps for planting a flower using order words. Your sequence is:

1. **First**, find a sunny spot in the garden.
2. **Next**, dig a small hole.
3. **Then**, place the flower seed in the hole.
4. **Finally**, water the soil.

This sequence is logical and clearly demonstrates your understanding of how to use order words to describe a process.

Great job! Are you ready to move on to the next part of the lesson, or would you like to practice more with another example?

Anonymous

no

ChatGPT

Alright, it seems like you have a good grasp of using order words now. Let's proceed to the next part of the lesson.

Part 4: Practice by giving some concrete story writing or planning tasks and provide evaluation and real-time feedback

Step 7: Now, I'd like you to describe the story of Snow White in a 5-step story using appropriate order words.

Remember, your story should use order words correctly, including both ordinal words (like first and second) and other order words (like next, then, after that, etc.). The story should follow a logical sequence and, if a specific number of steps is requested, should contain exactly that number.

Please describe the story of Snow White in five steps using order words.

Anonymous

First, Snow White go to trees. Next, Snow White find a small house. Then, the Queen find Snow White is alive.

ChatGPT

You've made a good start with the story of Snow White. Let's try to complete it with two more steps to make a total of five, and ensure we're using order words correctly.

Based on what you've provided:

1. **First**, Snow White goes to the forest.
2. **Next**, she finds a small house.
3. **Then**, the Queen discovers Snow White is alive.

Now, we need two more steps to complete the story. Here's one way to do it:

4. **After that**, Snow White eats a poisoned apple given by the Queen and falls into a deep sleep.
5. **Finally**, a prince comes, kisses her, and Snow White wakes up, leading to a happy ending.

This completes the story in five steps, using order words appropriately.

Would you like to try another story, or shall we move on to the next part of the lesson?

Anonymous

First, Snow White goes to the forest. Next, she finds a small house. Then, the Queen discovers Snow White is alive. after that, snow white eats a bad apple and sleep. finally, prince save prince snow.

ChatGPT

Great job! You've successfully completed the story of Snow White using order words in a logical sequence. Your story goes like this:

1. **First**, Snow White goes to the forest.
2. **Next**, she finds a small house.
3. **Then**, the Queen discovers Snow White is alive.
4. **After that**, Snow White eats a bad apple and falls asleep.
5. **Finally**, a prince saves Snow White.

This is a well-structured story using order words correctly. You've shown a good understanding of how to use these words to narrate a story in an ordered manner.

Now that you've completed the Snow White story successfully, you can move on to describing other stories or scenarios using order words. Would you like to try another story, or are you ready to move on to self-directed learning with topics of your choice?

Anonymous

yes

ChatGPT

Great! Let's try another story exercise. This time, you can choose one of the following scenarios to describe in a 5-step story using order words:

1. A field trip.
2. Cleaning the classroom at the end of the semester.
3. Preparing for the final exam.

Please select one scenario and describe it in five steps using order words like "first," "next," "then," "after that," and "finally."

Anonymous

first, choose a place. next, watch 氣象預報. then, buy food. after that, buy food. finally, catch a but.

ChatGPT

You've made a good effort in describing the scenario of a field trip using order words. Let's refine it a bit for clarity and to avoid repetition:

1. **First**, choose a place for the field trip.
2. **Next**, watch the weather forecast (氣象預報) to ensure good conditions.
3. **Then**, plan and buy food for the trip.
4. **After that**, gather all necessary supplies and equipment for the trip.
5. **Finally**, catch a bus to the chosen destination.

In this revised version, each step is clearly defined with an order word, and the actions are in a logical sequence. It looks like you're getting the hang of using order words effectively!

Would you like to try another scenario, or are you ready to move on to self-directed learning?

Anonymous

yes

ChatGPT

Fantastic! Let's try another scenario for practice. Please choose one of the following options and describe it in a 5-step story using order words:

1. Cleaning the classroom at the end of the semester.
2. Preparing for the final exam.

Select one and describe it in five steps using order words like "first," "next," "then," "after that," and "finally."

Anonymous

first, make a clean plan. next, clean classroom with friends. then, water flowers. after that, 丢垃圾.

Finally, put flower in classroom.

ChatGPT

You've done a great job describing the scenario of cleaning the classroom at the end of the semester using order words. Here's your story with a slight refinement for clarity:

1. **First**, make a cleaning plan.
2. **Next**, clean the classroom with friends.
3. **Then**, water the flowers.
4. **After that**, dispose of the trash (丢垃圾).
5. **Finally**, place fresh flowers in the classroom.

This sequence is well-organized and uses order words effectively to describe each step in the process. You've shown a good understanding of how to structure a task using order words.

Are you ready to move on to the next part of the lesson, or would you like to try another scenario?

Anonymous

yes.

ChatGPT

Great! Since you've successfully completed practicing with scenarios, let's move on to the next part of our lesson.

Part 5: Apply with learners' own chosen topics and provide evaluation and real-time feedback

Step 8: Self-directed Learning

Now it's time for you to choose a topic or activity that interests you. Describe it in a series of steps using order words. This is an opportunity for you to apply what you've learned in a context that's meaningful to you.

Please select a topic or activity of your choice and describe it in five steps using order words like "first," "next," "then," "after that," and "finally."

Anonymous

First, assign everyone's work. Second, pick a tool. Third, clean the floor. Then, collect trash. Finally, have a clean floor.

ChatGPT

You've done a wonderful job applying the concept of order words to your chosen activity. Here's how your sequence looks with a slight adjustment for clarity:

1. **First**, assign everyone's work for cleaning.
2. **Next**, pick the necessary tools for cleaning.
3. **Then**, clean the floor.
4. **After that**, collect all the trash.
5. **Finally**, enjoy the clean floor.

Your sequence is logical and well-structured, demonstrating a clear understanding of how to use order words effectively. You've successfully applied the concept to a practical scenario, which is excellent!

Would you like to try describing another activity, or are you ready to move on to the final part of our lesson?

Anonymous

move

ChatGPT

Great! Let's move on to the final part of our lesson.

Part 6: Report student's overall performance and suggest for further study

Step 9: Provide an overall remark about student's learning performance

Throughout this lesson, you've shown a clear understanding of the concept of order words. You've successfully applied these words in various scenarios, from the life cycle of a butterfly to practical activities like planning a field trip and cleaning a classroom. Your ability to structure information sequentially using order words like "first," "next," "then," "after that," and "finally" has improved significantly.

For further study, I recommend practicing with more complex scenarios or trying to use order words in your daily conversations and writings. This practice will help solidify your understanding and make your communication clearer and more effective.

Step 10: Congratulations and Goodbye!

Congratulations on mastering the usage of order words! Your progress is commendable, and these skills will be very useful in many aspects of communication. Keep practicing, and don't hesitate to explore more advanced aspects of language learning.

Goodbye, and well done!